Compiler construction lab

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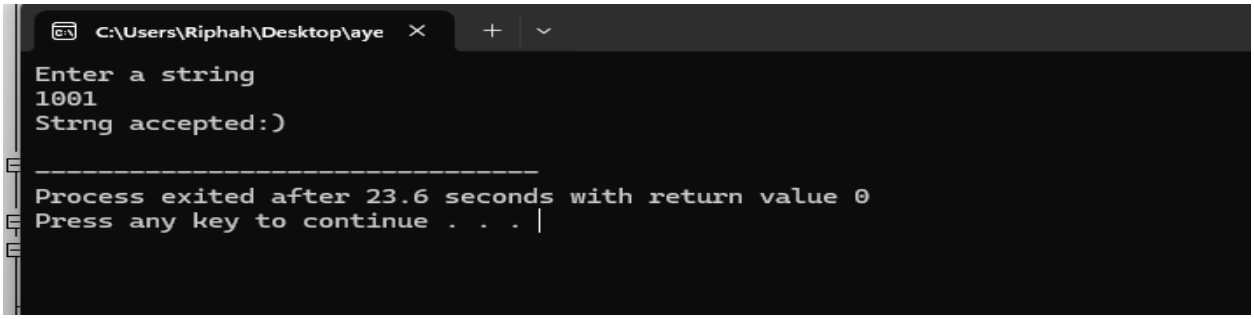
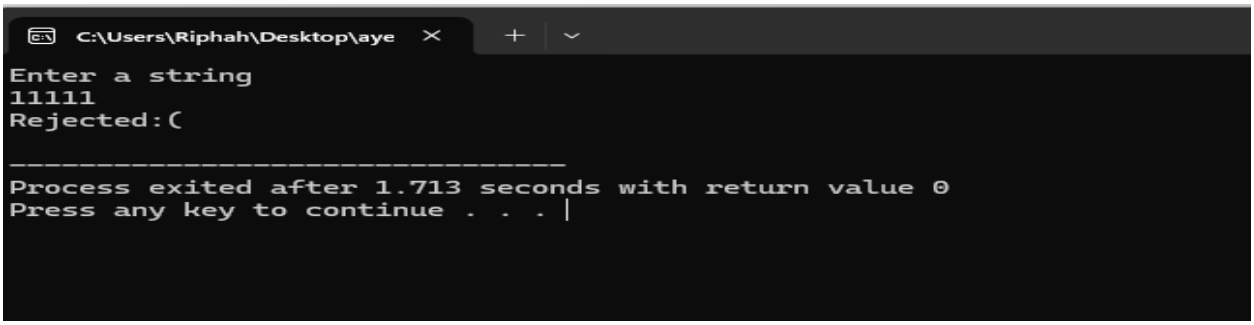
Task1:

```
#include<iostream>
using namespace std;
int main()
{
	string input;
	cout<<"Enter a string"<<endl;
	cin>>input;

	int state=0;
	for (char c:input){
		if(state==0)
		{
			if(c=='1'){
				state=0;
			}
			else {
				state=1;
			}

		}
			else
			{
			if(c=='0' || c=='1')
			{
				state=1;
			}
```

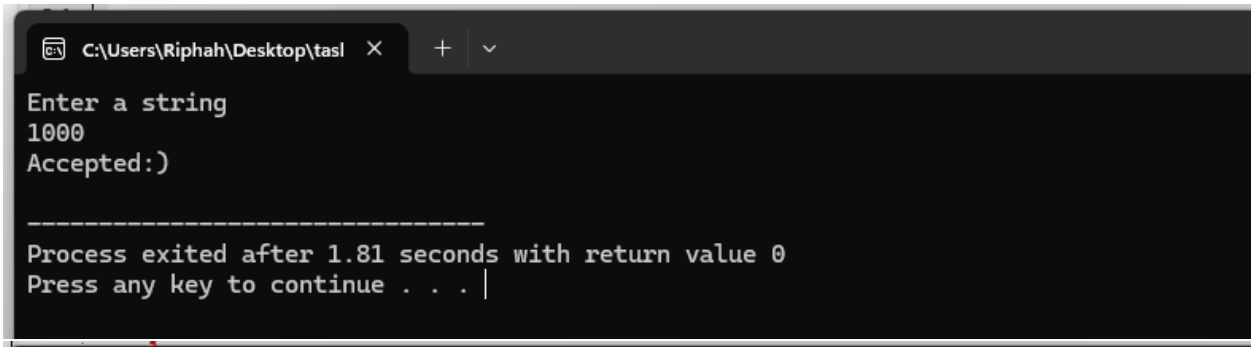
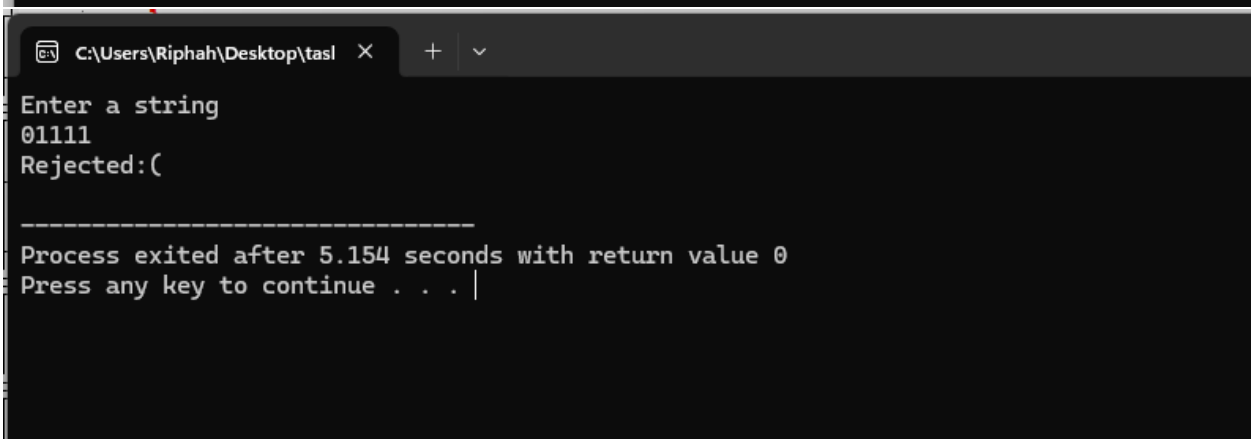
```
            }

        }
            if(state==1)
        {
            cout<<"Strng accepted:)"<<endl;
        }
        else
        {
            cout<<"Rejected:("<<endl;
        }

}
```





Task2:

```
#include<iostream>
using namespace std;
int main()
{
	string input;
	cout<<"Enter a string"<<endl;
	cin>>input;

	int state=0;
	for (char c:input){
	if(state==0){

	if(c=='1')

	{
		state=1;
	}
	else
	{
		state=5;
	}
	}
	else if(state==1){

	if(c=='0')
		{
			state=2;
		}
```

```
                else{
                        state=1;
                }

        }
                else if(state==2){

        if(c=='0')
                {
                        state=2;
                }
                else{
                        state=1;
                }

        }
        else{
                if(c=='0'||c=='1')
                {
                        state=5;1
                }
        }}

if(state==2)
{
        cout<<"Accepted:)"<<endl;

}
else{
```

```
        cout<<"Rejected:("<<endl;
}
}
```